

Lecture 10: Classification metrics

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Machine Learning I: 46-927

December 2, 2020

Announcements

- ▶ Homework was just due. There will be a quiz on Friday. Everything is back to the usual schedule.

Today

Today we will talk more about classification metrics, and then potentially introduce generative models.

- ▶ Talk about the homework
- ▶ Classification:
 - ▶ Imbalanced data
 - ▶ Classification metrics
- ▶ Maybe a bit of decision trees

Classification so far

Focus on binary classification,

- ▶ $Y = 1$ if an event of interest happened
- ▶ $Y = 0$ if it did not happen

Most of our classifiers will give us a guess $\hat{p}(x) = \hat{\mathbb{P}}(Y = 1|X = x)$.
This is an estimate of the true $p(x) = \mathbb{P}(Y = 1|X = x)$.

We saw that to minimize average misclassification error,
 $\mathbb{E}\mathbf{1}_{f(X) \neq Y}$, you should choose $f(x) = \mathbf{1}_{p(x) > 1/2}$.

Let's think about this more carefully.

Misclassification rate

Suppose that I tell you my classifier has 97% accuracy. Is this good?

Misclassification rate

Suppose that I tell you my classifier has 97% accuracy. Is this good?

It depends! What if I am predicting lottery wins. The probability of winning may be $1/175000000$.

In that case, guessing "no" has a misclassification rate of 5×10^{-9} ...

We need to consider the base rate: the probability of each class given no further information.

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Confusion matrix: Example

For the homework problem (to be), $Y = 1$ if an individual responds to marketing. The classification rule $\hat{p}(x) > 0.5$ gives a confusion matrix similar to:

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	0	2
Guess $Y = 0$	1083	7957

Issues with imbalanced data and misclassification rates

The idea of minimizing the misclassification rate is tempting — no one wants to be wrong!

However, this is often not the real goal of applying a classifier. Losses are usually asymmetric! Sometimes you even care directly about $\hat{p}(x)$!

Furthermore, in settings where the classes are imbalanced, the misclassification rate is very difficult to improve and is often misleading.

Imbalanced data

In many common applications, our classes are imbalanced:

- ▶ Marketing
- ▶ Fraud detection
- ▶ Mortgage default
- ▶ Whether there will be a large jump in price after an event.

Misclassification rate and imbalanced data

We know that we should classify as $Y = 1$ when $P(Y = 1|X = x) > 0.5$ to minimize expected misclassification rate. Suppose that an event only happens 5% of the time in the general population.

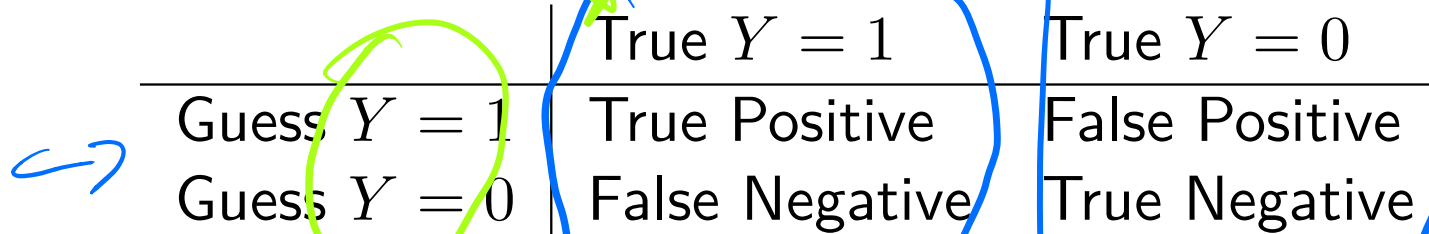
By Bayes Theorem,

$$P(Y = 1|X = x) = \frac{P(Y = 1) P(X = x|Y = 1)}{P(X = x)}$$

Therefore, to classify as $Y = 1$, we would need to see an X that is 10 more likely in the $Y = 1$ population than the general population.

Even if there is useful information in the data, it is probably not enough to flip our classifications!

Confusion matrix: error rates



The confusion matrix is presented with handwritten annotations. A green circle highlights the 'Guess Y = 0' row. A blue circle highlights the 'True Y = 1' column. A blue box highlights the 'True Y = 0' column. Arrows point from the title to the matrix components.

	True Y = 1	True Y = 0
Guess Y = 1	True Positive	False Positive
Guess Y = 0	False Negative	True Negative

Other error rate measures better capture asymmetric concerns when classifying:

► Sensitivity (Recall, Hit rate):

$$\frac{TP}{TP + FN} = P(\text{guess } 1 \mid \text{true } 1)$$

► Specificity (True Negative Rate, TNR):

$$\frac{TN}{TN + FP} = P(\text{guess } 0 \mid \text{true } 0)$$

These quantities will allow us to better think about cost trade offs.

Sensitivity and Specificity

Sensitivity: Fraction of points with $Y = 1$ that we find.

- ▶ Want high Sensitivity when False Negatives are more costly than False Positives. Example: Fraud detection.

Specificity: Fraction of points with $Y = 0$ that we avoid.

- ▶ Want high Specificity when False Positives are more expensive than False Negatives. Example: Criminal trials.

We would like both of them to be high, but we need to make tradeoffs as we adjust our threshold.

- ▶ We can flag more cases as fraud to improve sensitivity, but this will decrease our specificity.

$$P(Y=1 | X=x) > c$$

Cost/Loss for classification

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	0	$L_{01} = 5$
Guess $Y = 0$	$L_{10} = 100$	0

You will show on your homework that the loss is minimized if you guess $Y = 1$ when

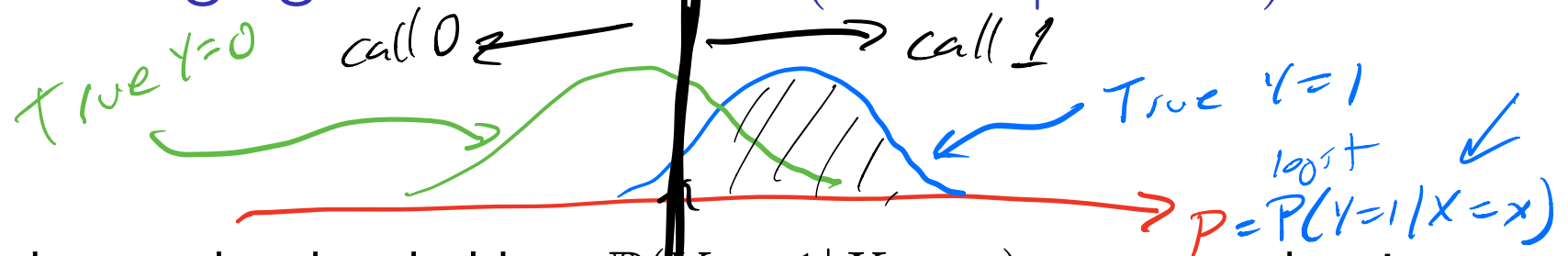
$$\mathbb{P}(Y = 1|X = x) > \frac{L_{01}}{L_{01} + L_{10}} = \frac{5}{100 + 5} \approx 5\%$$

If it costs \$5 dollars to call someone who will not buy, but it costs \$100 to miss someone who would buy, you should

call it $\mathbb{P}(Y=1|X=x) > 5\%$

This suggests that we can trade off costs and error rates by just changing the threshold on $\mathbb{P}(Y = 1|X = x)$!

Changing threshold on $\mathbb{P}(Y = 1|X = x)$



As we change the threshold on $\mathbb{P}(Y = 1|X = x)$, we vary the size and quality of the set we reject.

We can think about $\mathbb{P}(Y = 1|X = x)$ as a function for ranking our observations by likelihood of $Y = 1$.

It is convenient to look at the Sensitivity and Specificity of our classifier as the threshold changes. This corresponds to the ROC (Receiver operating characteristic) curve that you will plot in your homework.

Aside: Classification scores

So far, we've been talking about classification based on estimates

$$\hat{p}(x) = \mathbb{P}(Y = 1 | X = x).$$

logit $\hat{p}(x)$ ←

In fact, we don't really need probabilities to make classifications. A more general concept is a classification score, $\hat{f}(x)$, such that $\hat{f}(x)$ is big for

Any threshold on $\hat{f}(x)$ will yield a confusion matrix, which in turn yields all of the error metrics we have discussed, as well as ROC curves.

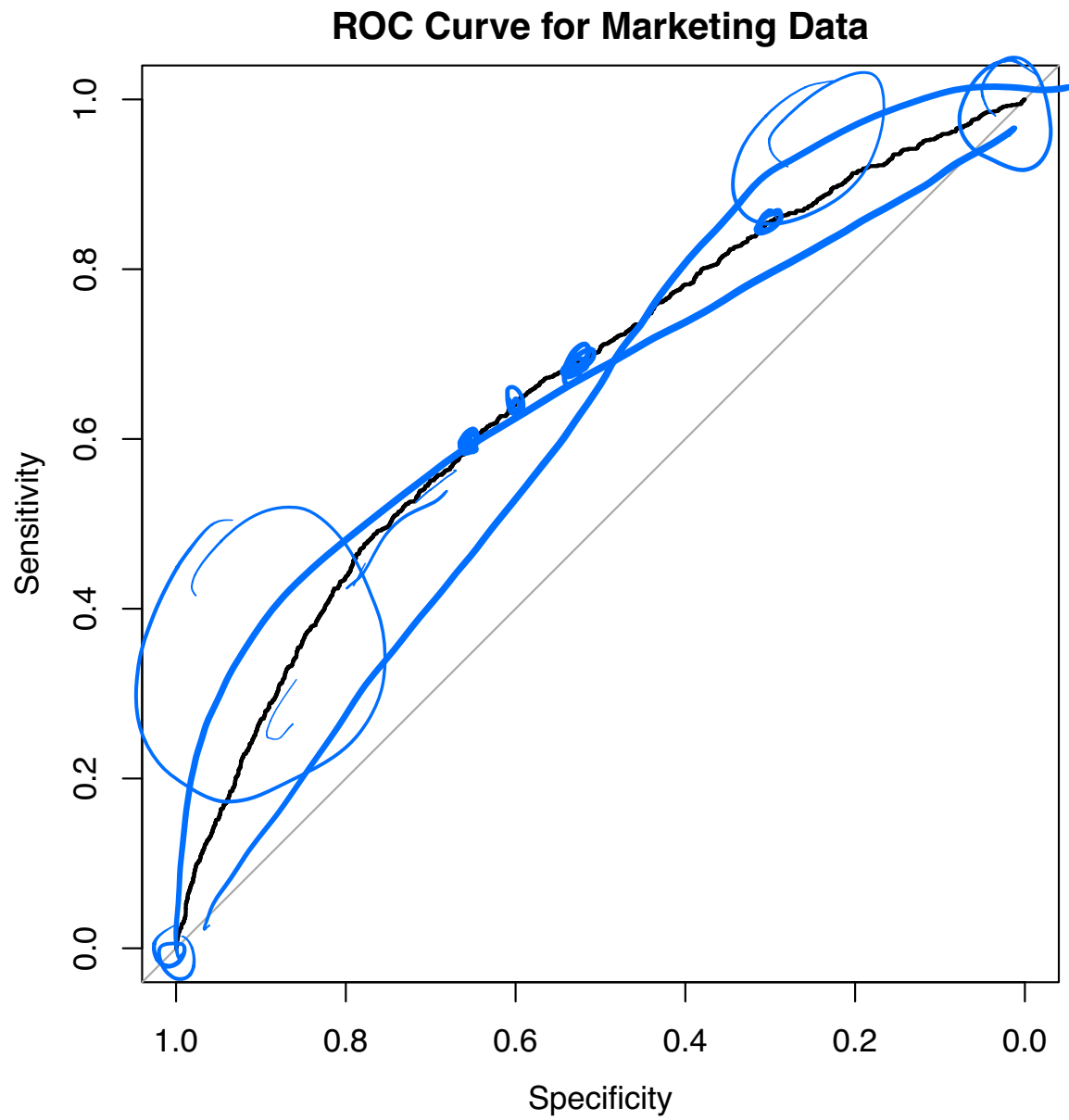
- Question: How do ROC curves change if I monotonically warp my classification score?

Aside: Classification scores

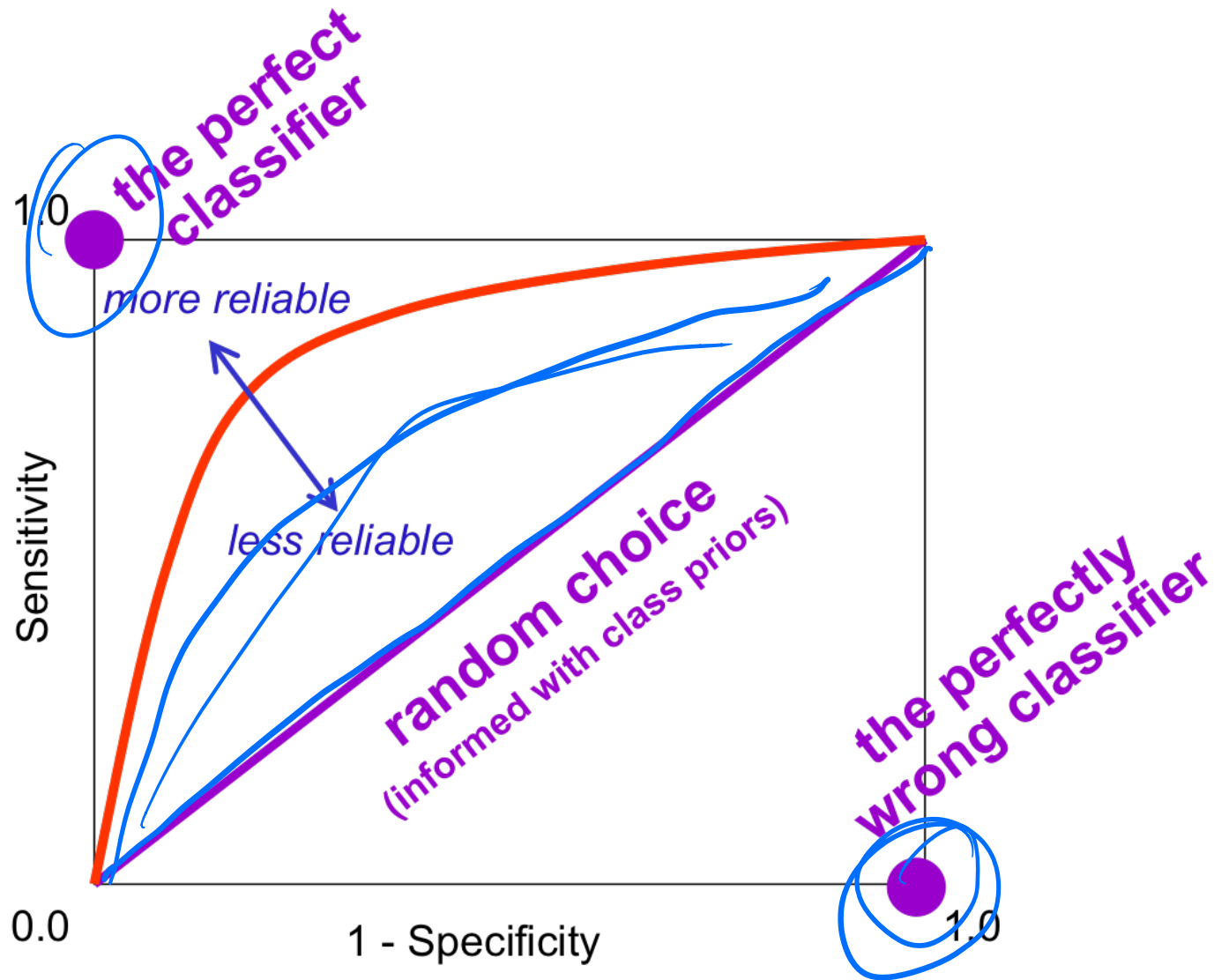
Examples of classification scores that are not probabilities:

It is often useful to think of the distribution of $\hat{f}(x)|Y = k$ for understanding classifier behavior. (Homework)

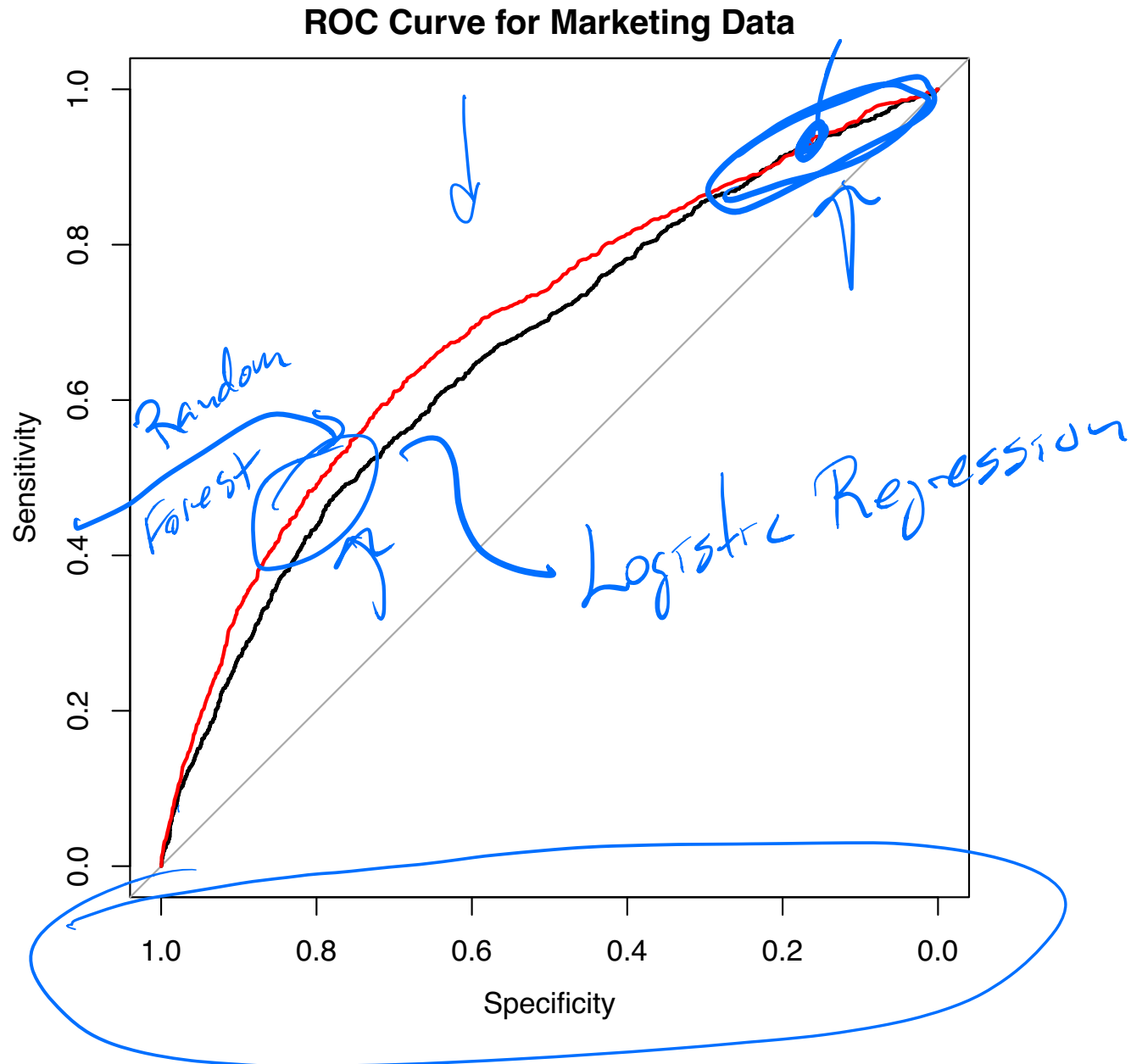
ROC Curve



ROC Curve



ROC Curve (again)



ROC Curve

A generally higher curve tends to be better, though you should keep tradeoffs in mind.

The Area Under the Curve (AUC) is one way to measure this. A higher area tends to be more attractive.

AUC has a nice interpretation:

If you randomly pick two observations from your distribution and order them by $\hat{p}(x)$, the AUC is the probability that your ordering is correct!

In reality though, you are interested in the performance cutoff at a particular point on the curve